ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Cornerstone project work entitled "LIBRARY MANAGEMENT SYSTEM" is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2025-2026.

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DECLARATION

We hereby declare that the Cornerstone project entitled "**LIBRARY MANAGEMENT SYSTEM**" is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree** in **Computer Science and Engineering**. We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy

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ABSTRACT

The Library Management System (LMS) is a desktop-based application designed to efficiently manage library records by replacing traditional manual methods with a digital solution. Thesystem enables users to perform operations such as adding, updating, deleting, issuing, and searching book and member information in a structured and secure manner using a relational database. Developed using Core Java, Swing for the graphical user interface, JDBC for database connectivity, and Oracle Database for backend storage, the application ensures data accuracy, reduces redundancy, and improves accessibility. With its user-friendly interface and efficient data handling capabilities, the system enhances institutional productivity and provides a reliable solution for managing library resources.

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ACRONYMS

- **LMS** : Library Management System
- **JDBC** : Java Database Connectivity
- **GUI** : Graphical User Interface
- **SQL** : Structured Query Language

1.INTRODUCTION

1.1 Brief Information about the Project

The Library Management System (LMS) is a desktop-based application developed to efficiently manage library operations within an organization. It provides a centralized platform to store, retrieve, update, and manage information related to books, members, and transactions such as issuing and returning books.

This system replaces traditional manual record-keeping methods with a computerized solution, thereby improving data accuracy, reducing redundancy, and enhancing accessibility. The application is developed using Core Java for programming, Swing for designing the graphical user interface, JDBC for database connectivity, and Oracle Database for storing library data.

The system is designed to be user-friendly and suitable for use in schools, colleges, and public libraries.

1.2 Motivation and Contribution of the Project

In many libraries, records are still maintained manually using registers or basic spreadsheets. This leads to several issues such as difficulty in tracking books, managing member records, and handling issue/return processes efficiently.

These challenges motivated the development of the Library Management System.

The key contributions of this project include:

- Providing a digital solution for managing library records
- Reducing manual effort and paperwork
- Improving accuracy and consistency of data
- Enabling quick search and retrieval of books and member information
- Simplifying book issue and return processes
- Enhancing data security through controlled access

This system improves the efficiency and management of library operations.

1.3 Objectives of the Project

The main objectives of the Library Management System are:

- To develop a user-friendly application for managing library data
- To maintain records of books and members in a structured database

- To provide functionalities for issuing and returning books
- To enable quick searching of books using title, author, or ID
- To track availability status of books
- To display library records in an organized format

1.4 Scope of the Project

The Library Management System can be used in schools, colleges, and small to medium-sized libraries to manage their operations efficiently.

The system supports the following operations:

- Adding new books and member details
- Updating existing records
- Deleting book or member information
- Issuing and returning books
- Searching books and members using various criteria
- Displaying all records in an organized manner

2. SYSTEM ANALYSIS

2.1 Functional Requirements

Functional requirements define the core operations and functionalities that the Library Management System (LMS) must perform to effectively manage library-related information.

The functional requirements of Library Management System include:

- **Add Book Details**
  The system should allow the administrator to add new book records, including details such as book ID, title, author, publisher, category, and availability status. Proper validation should be applied to ensure data accuracy.
- **Update Book Records**
  The system should provide functionality to modify or update existing book information such as title, author, or availability status.
- **Delete Book Information**
  The system should allow authorized users to delete book records. A confirmation mechanism should be included to prevent accidental deletion.
- **Search Book**
  The system should support searching for books using criteria such as book ID, title, or author, and return accurate results quickly.
- **Display Book Records**
  The system should display all book records in a structured format with options for sorting and filtering.
- **Manage Member Details**
  The system should allow adding, updating, and deleting member records. Each member should have unique identification.
- **Issue Book**
  The system should allow issuing books to members and record the issue date.
- **Return Book**
  The system should manage book returns and update availability status accordingly.

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

- **Performance**
  The system should respond quickly to operations like search, issue, and return.

- **Security**
  Only authorized users should access and modify data through authentication.
- **Usability**
  The system should be simple and easy to use.
- **Reliability**
  The system should work consistently without data loss.
- **Scalability**
  The system should handle increasing number of books and users efficiently.

2.3 Requirements Specification

2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 20 GB free space
- Monitor: Standard display
- Keyboard and Mouse

2.3.2 Software Requirements

- Operating System: Windows 7/10/11
- Programming Language: Java
- IDE: VS Code
- Database: Oracle Database
- Connectivity: JDBC

3. TECHNOLOGY DESCRIPTION

3.1 Programming Language

3.1.1 Introduction to Java

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is widely used for building desktop and enterprise applications.

In the Library Management System, Java is used to implement core functionalities such as adding books, issuing books, returning books, and searching records. Its features like platform independence, exception handling, and strong memory management help in building reliable applications.

3.2 Framework / Libraries

3.2.1 Introduction to Swing

Swing is used to create graphical user interfaces (GUI) in Java applications.

In this project, Swing is used to design interfaces such as:

- Login Window
- Add Book Form
- Issue/Return Book Window
- Search Book Window
- Display Records

Swing improves user interaction and makes the application easy to use.

3.3 Database Technology

The Library Management System uses a relational database to store book and member data in a structured manner. Oracle Database is used as backend, and JDBC is used for connectivity.

3.3.1 JDBC (Java Database Connectivity)

JDBC is used to connect Java applications with databases.

It performs:

- Database connection
- Executing SQL queries (INSERT, UPDATE, DELETE, SELECT)

- Retrieving data
- Closing connections

3.3.2 Oracle Database

Oracle Database is used to store data such as book details, member information, and transaction records. It ensures data integrity and efficient data handling.

3.4 Tools & IDEs

3.4.1 Visual Studio Code

VS Code is used as the development environment. It provides features like syntax highlighting, debugging, and code completion.

4. DATABASE DESIGN

4.1 Database Tables

The system uses two main tables:

- Book Table → stores book details
- Member Table → stores member information

4.2 Table Structure

4.2.1 Book Table

Field Name	Data Type	Description	Constraint
book_id	NUMBER	Book ID	Primary Key
title	VARCHAR2(100)	Book Title	Not Null
author	VARCHAR2(100)	Author Name	-
publisher	VARCHAR2(100)	Publisher	-
category	VARCHAR2(50)	Book Category	-
status	VARCHAR2(20)	Availability	-

4.2.2 Member Table

Field Name	Data Type	Description	Constraint
member_id	NUMBER	Member ID	Primary Key
name	VARCHAR2(100)	Member Name	Not Null
phone	VARCHAR2(15)	Contact	-
email	VARCHAR2(100)	Email	-

4.3 Keys and Constraints

- Primary Keys:
  - Book_ID in Book table
  - Member_ID in Member table
- **NOT NULL Constraint:**
  - **Important fields like book title and member name**
- **Data Type Constraints:**
  - **Ensures valid data types**

4.4 Table Relationships

- **One member can issue multiple books**
- **Each book can be issued to one member at a time**

This represents a one-to-many relationship.

5. SYSTEM DESIGN

5.1 System Architecture

The system follows a three-layer architecture:

1. **Presentation Layer**
   - Built using Java Swing
   - Handles user interaction
2. **Application Layer**
   - Processes logic (issue, return, search)
3. **Data Layer**
   - Stores data in SQL Database

5.2 Modules Description

1. Book Management Module

- Add books
- Update book details
- Delete books
- Display books

2. Member Module

- Add member details
- Manage member records

3. Issue/Return Module

- Issue books
- Return books
- Track availability

4. Search Module

- Search books by ID, title, or author

5. Database Connectivity Module

- Connects Java with Oracle DB

- Executes SQL queries

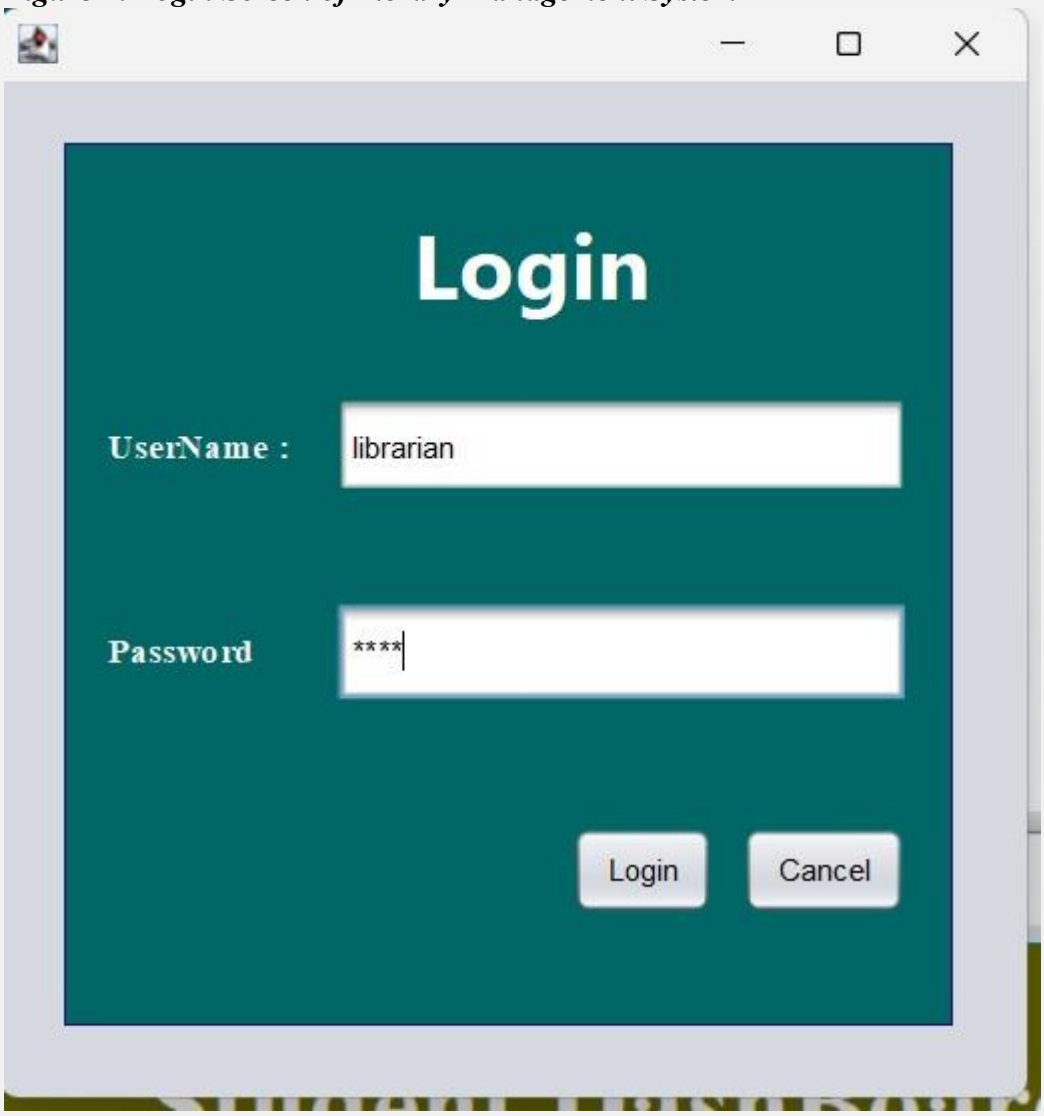
6. User Interface Module

- Provides GUI using Swing
- Handles user interaction

6.SYSTEM IMPLEMENTATION

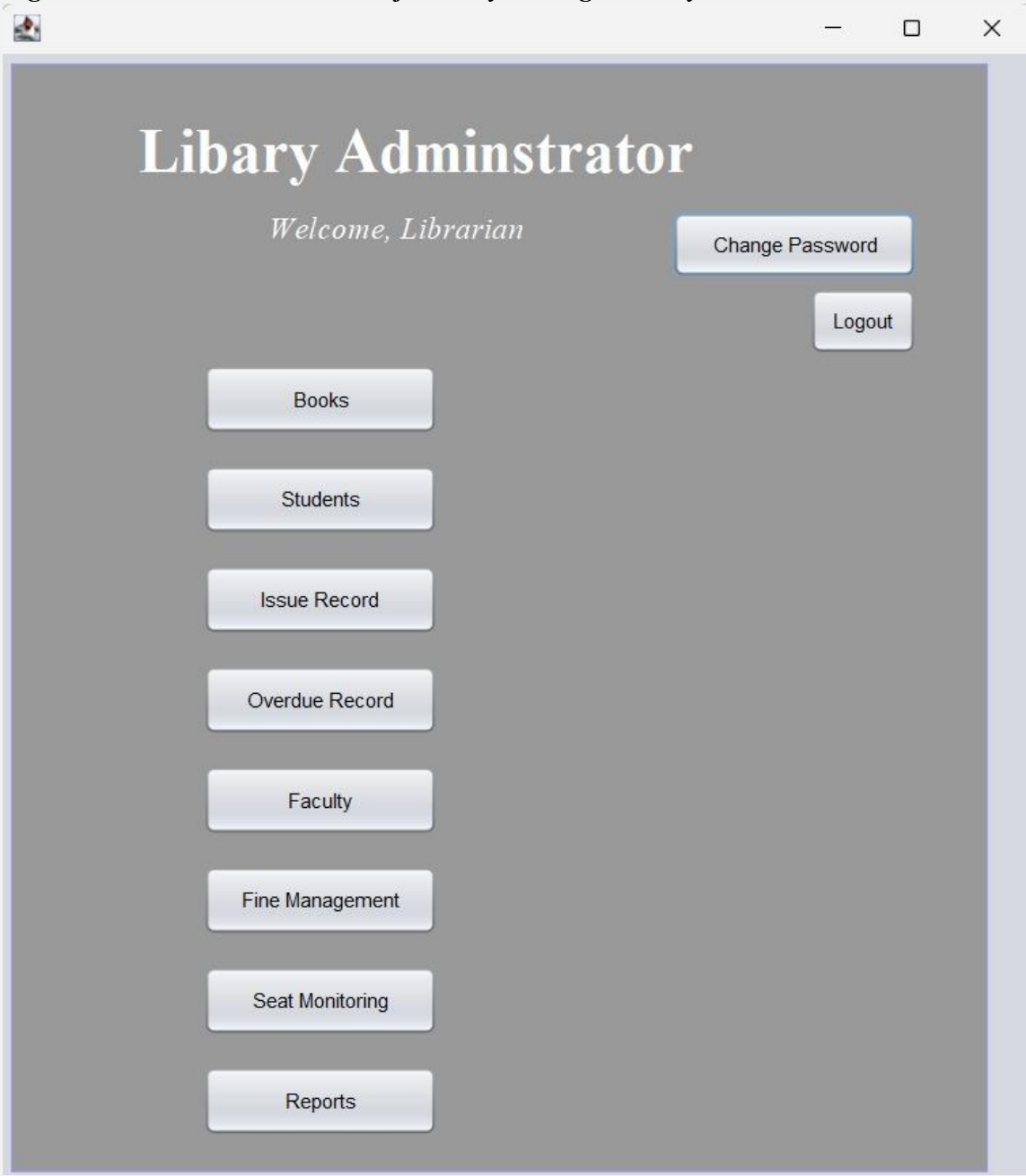
1. Login Screen

Figure 1: Login Screen of Library Management System



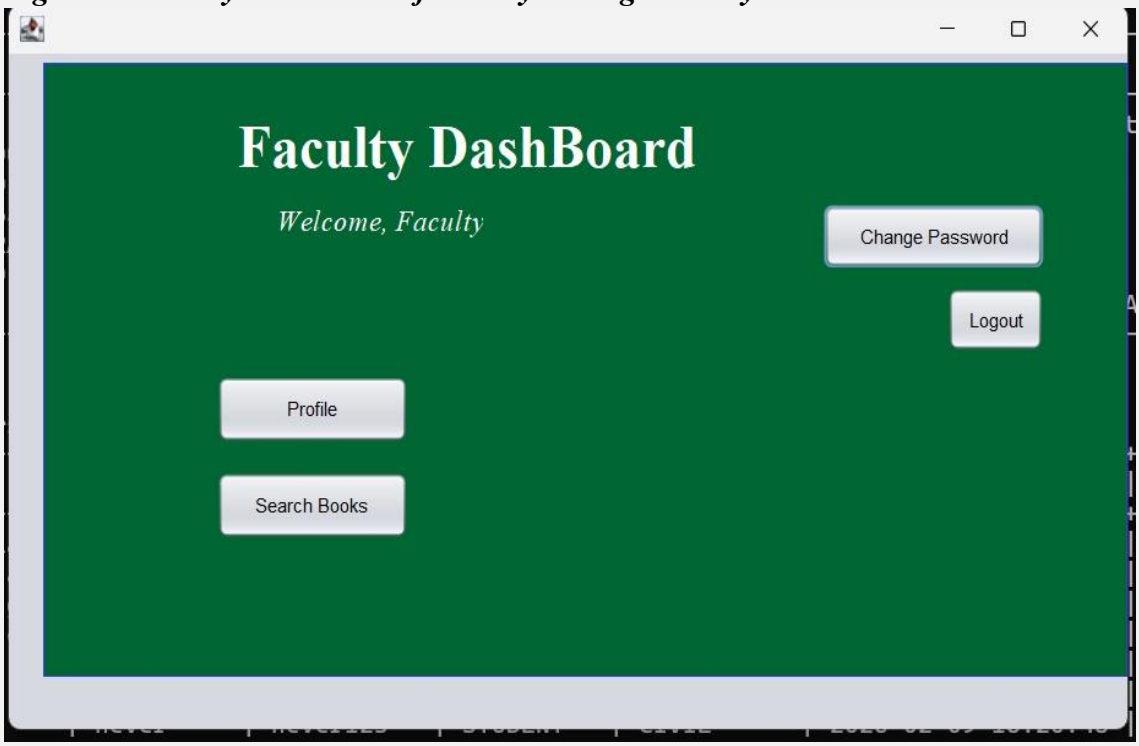
2. Librarian Dashboard

Figure 2: Librarian Dashboard of Library Management System



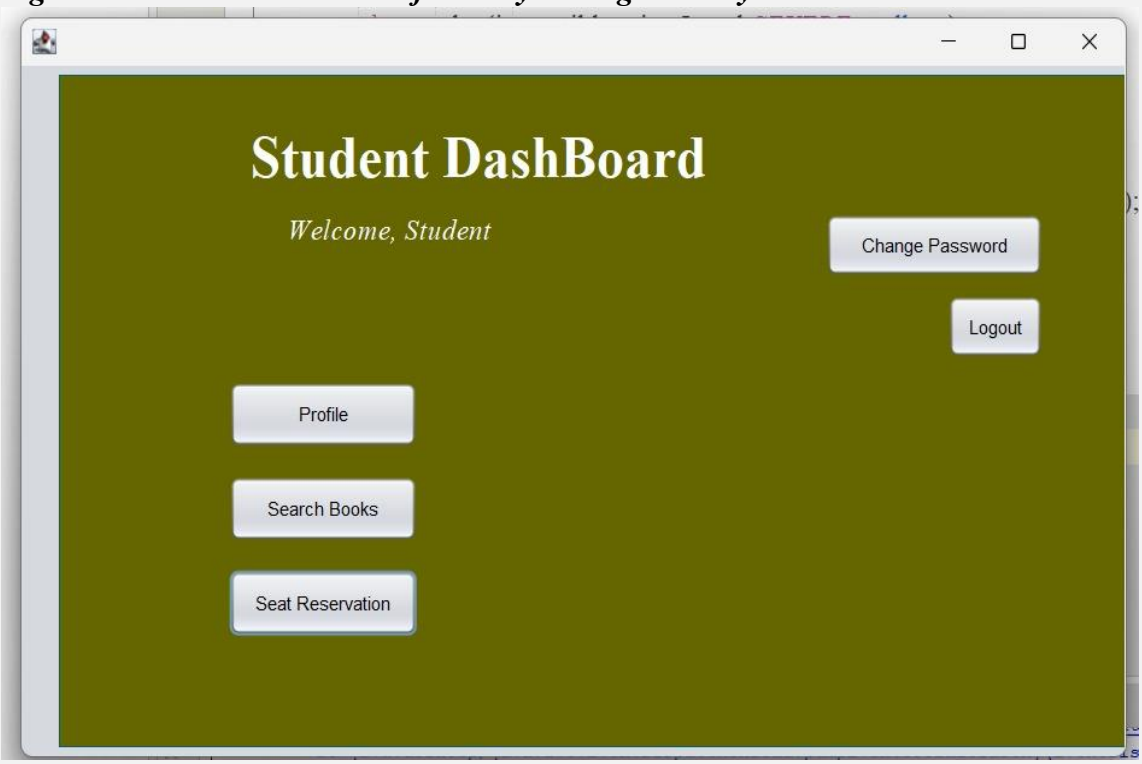
3. Faculty Dashboard

Figure 3: Faculty Dashboard of Library Management System



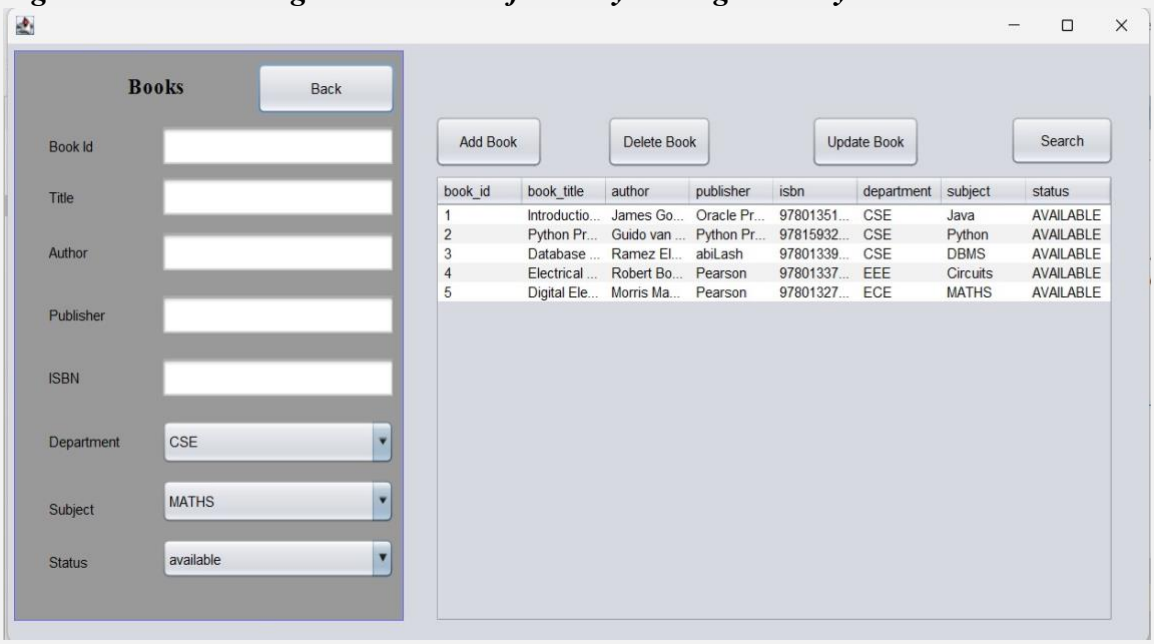
4. Student Dashboard

Figure 4: Student Dashboard of Library Management System



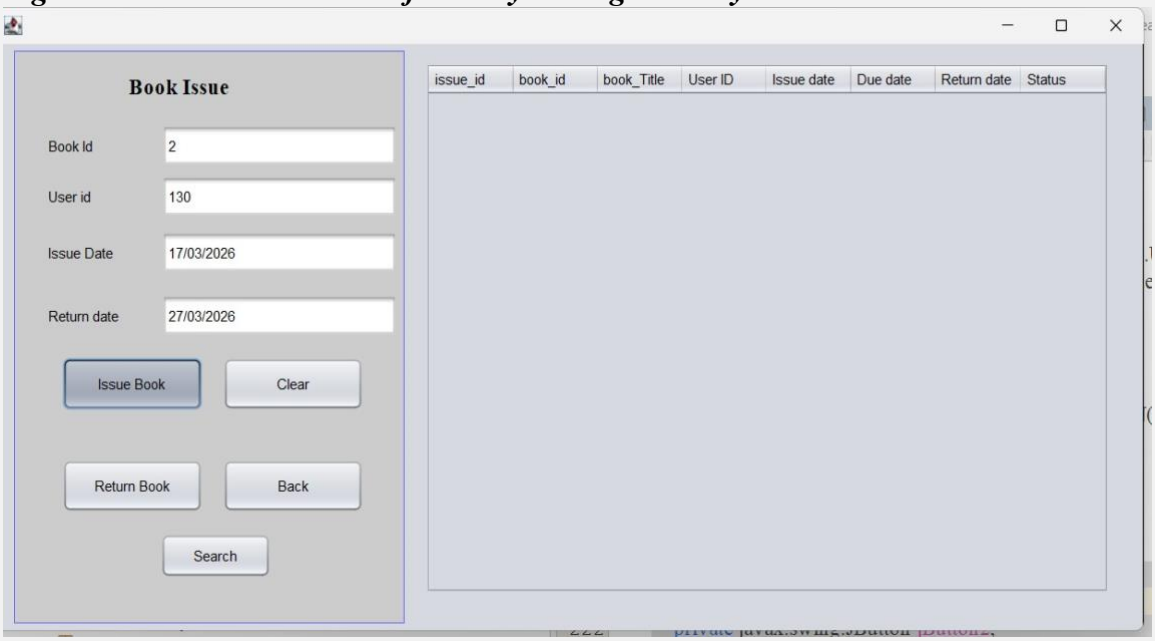
5. Book Management Screen

Figure 5: Book Management Screen of Library Management System



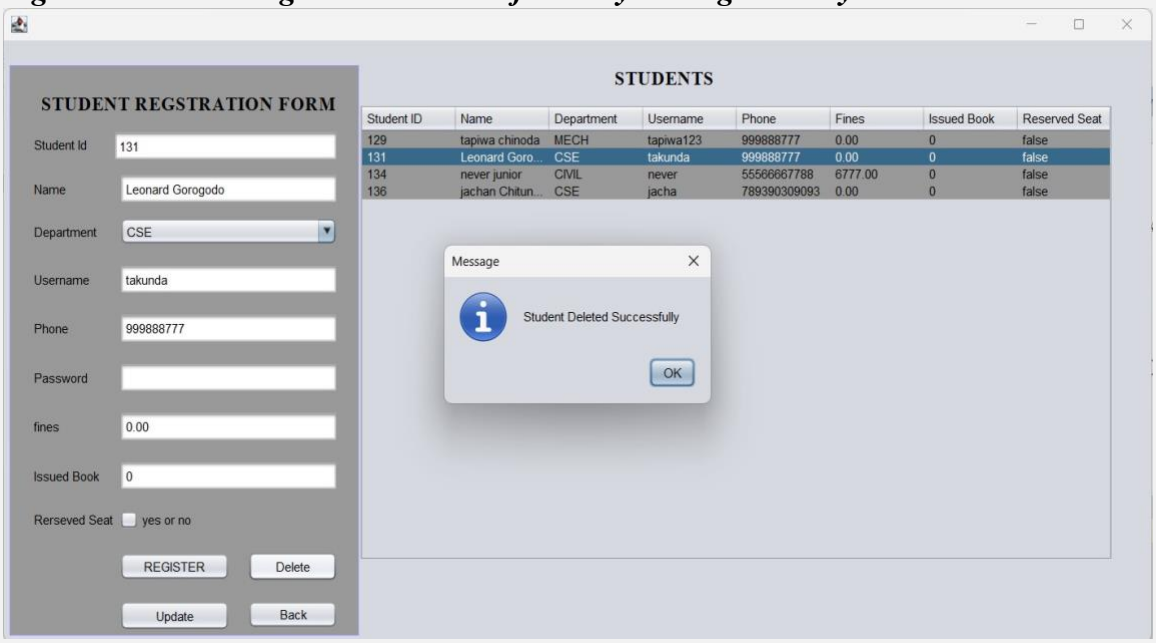
6. Book Issue Screen

Figure 6: Book Issue Screen of Library Management System



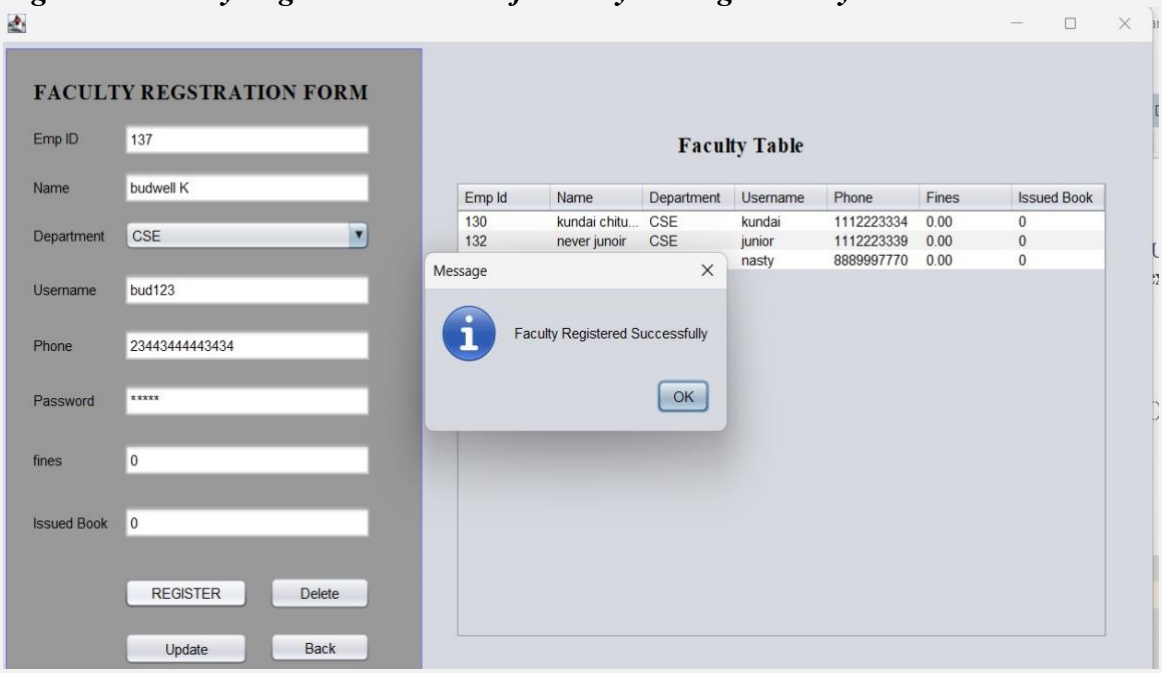
7. Student Registration Form

Figure 7: Student Registration Form of Library Management System



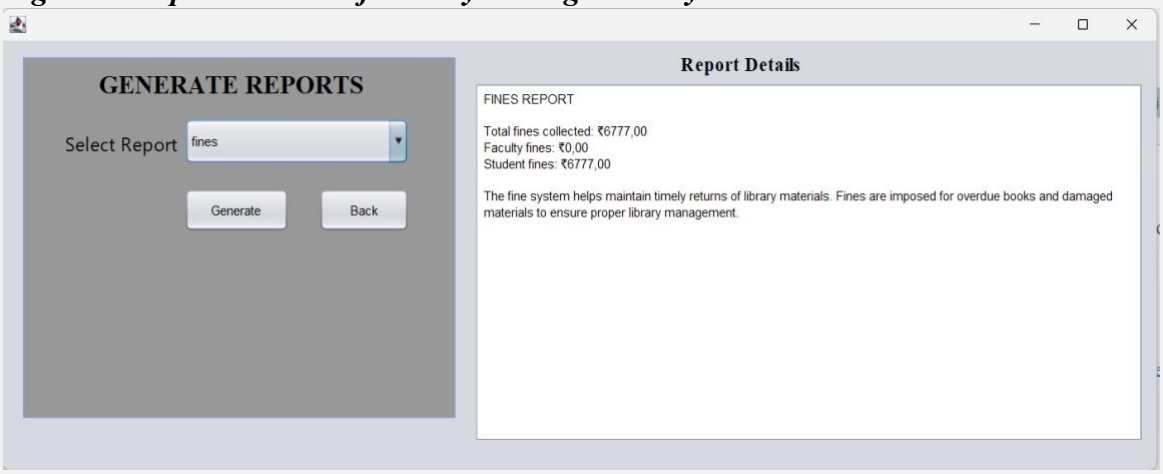
8. Faculty Registration Form

Figure 8: Faculty Registration Form of Library Management System



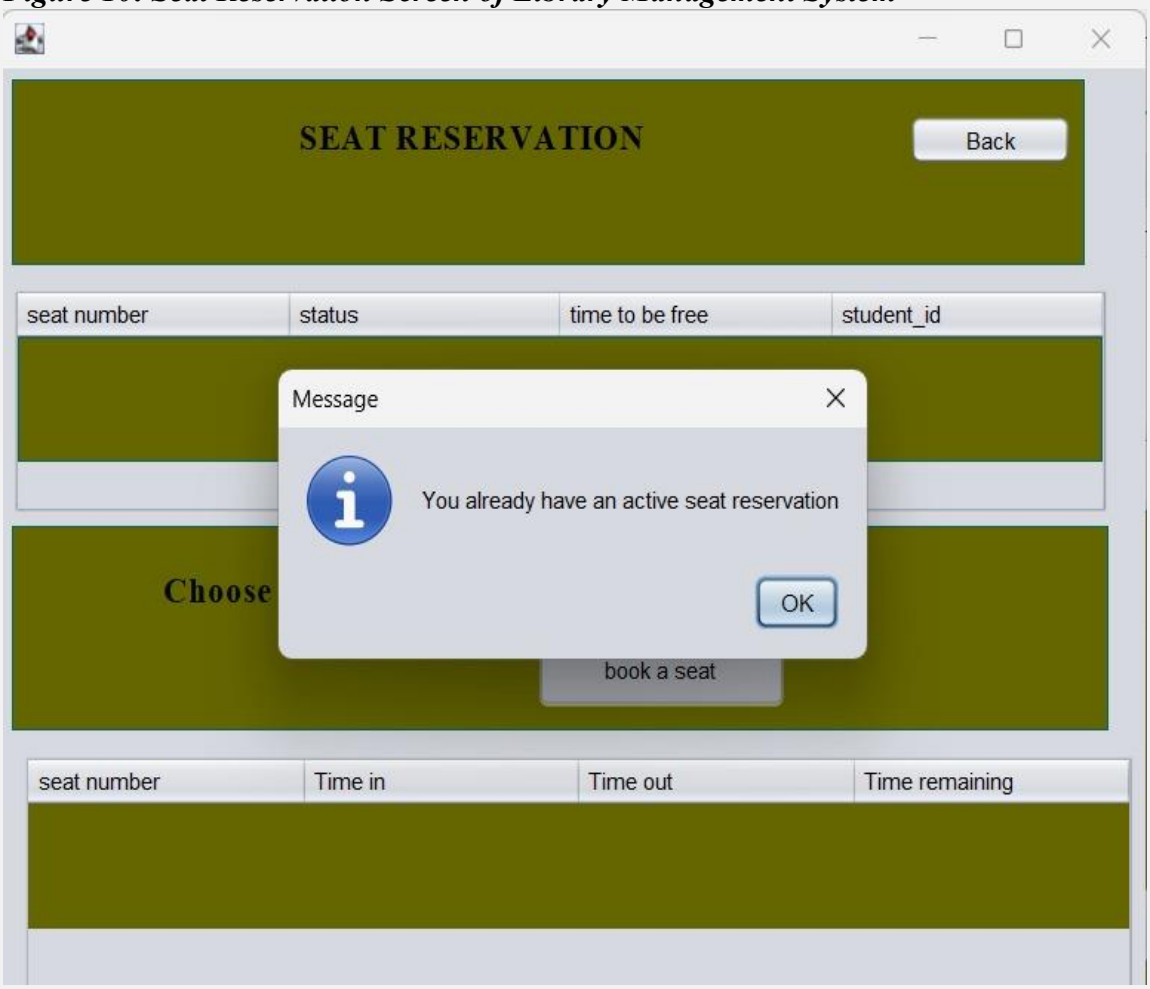
9. Reports Screen

Figure 9: Reports Screen of Library Management System



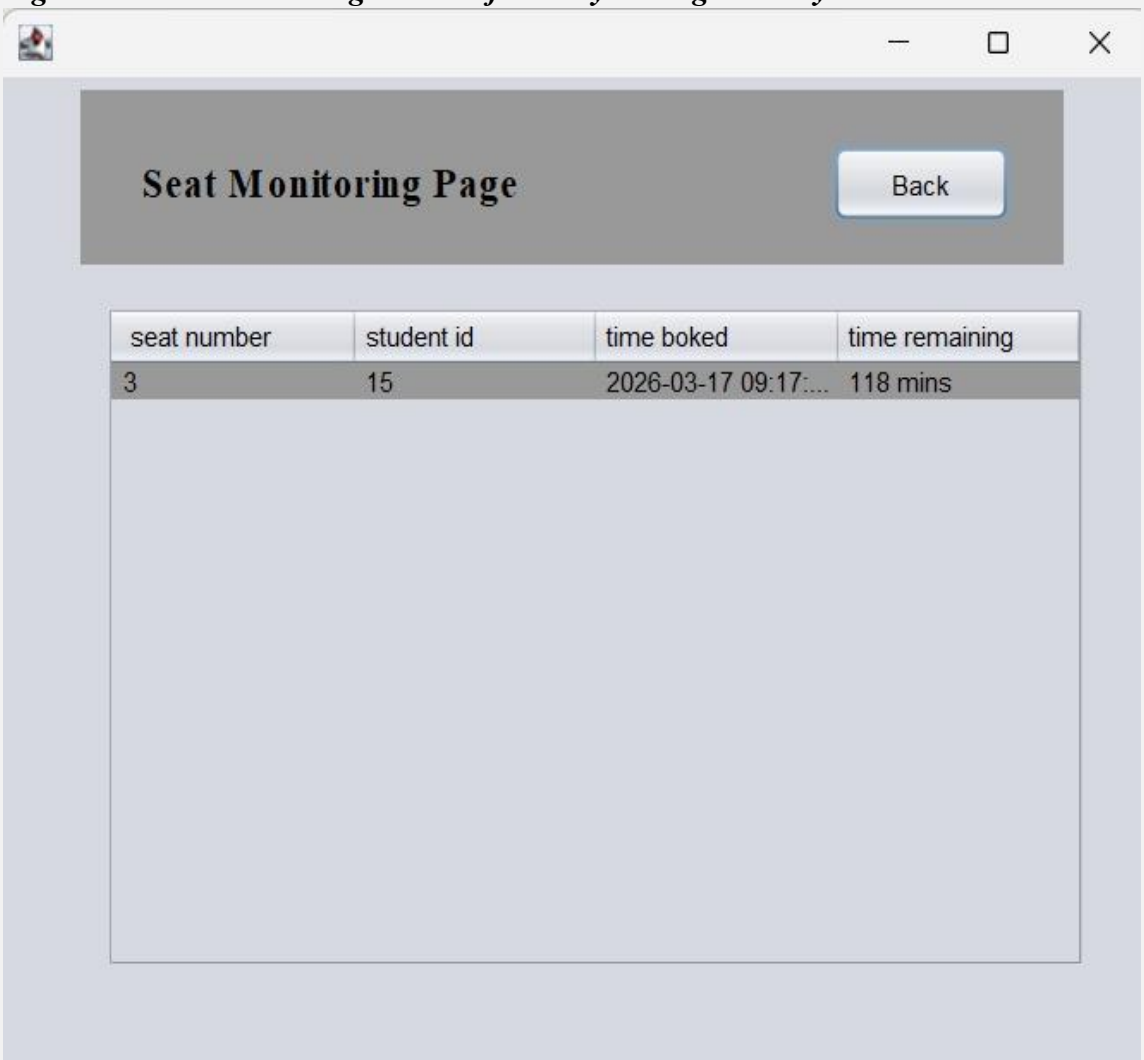
10. Seat Reservation Screen

Figure 10: Seat Reservation Screen of Library Management System



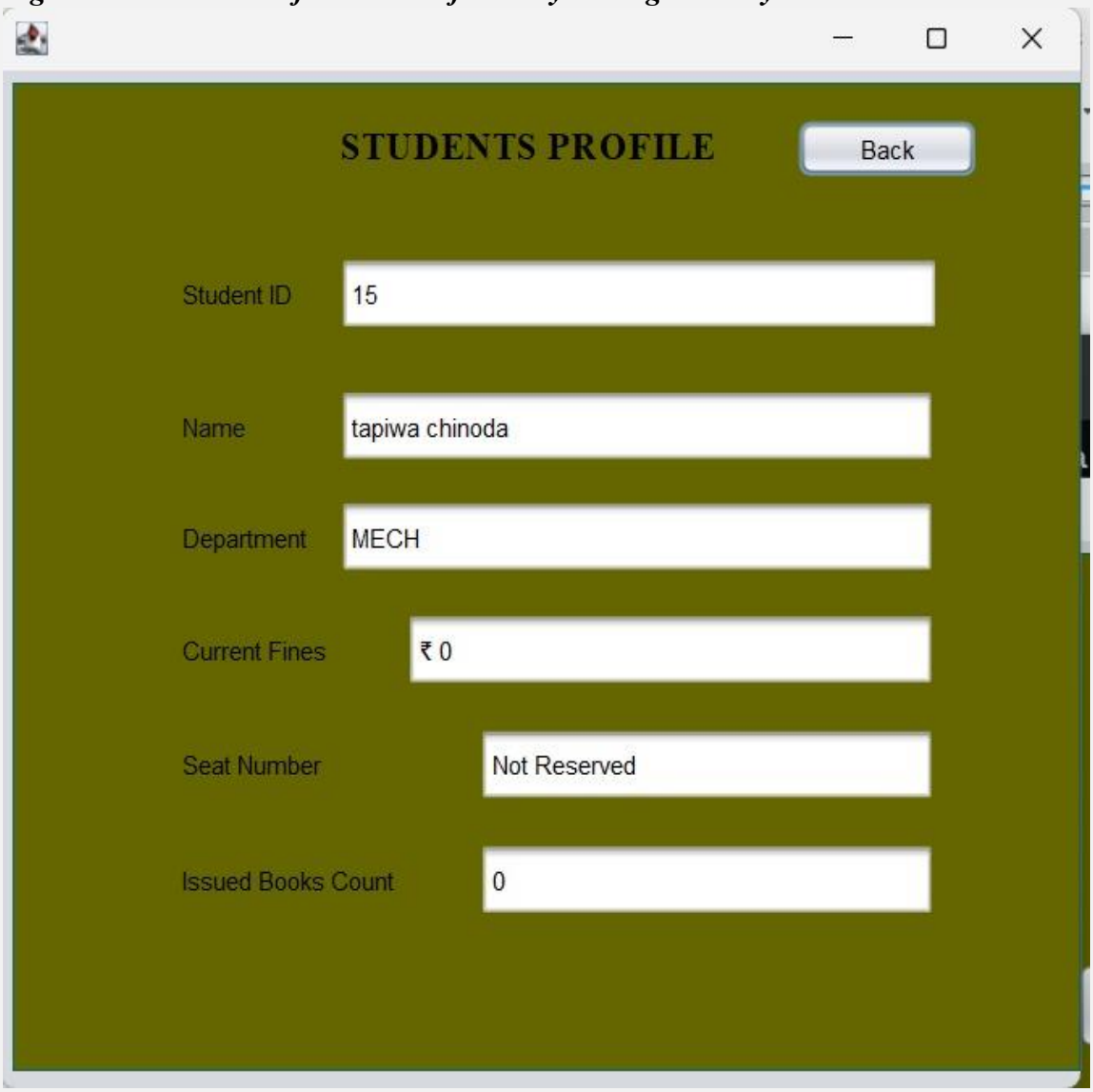
11. Seat Monitoring Screen

Figure 11: Seat Monitoring Screen of Library Management System



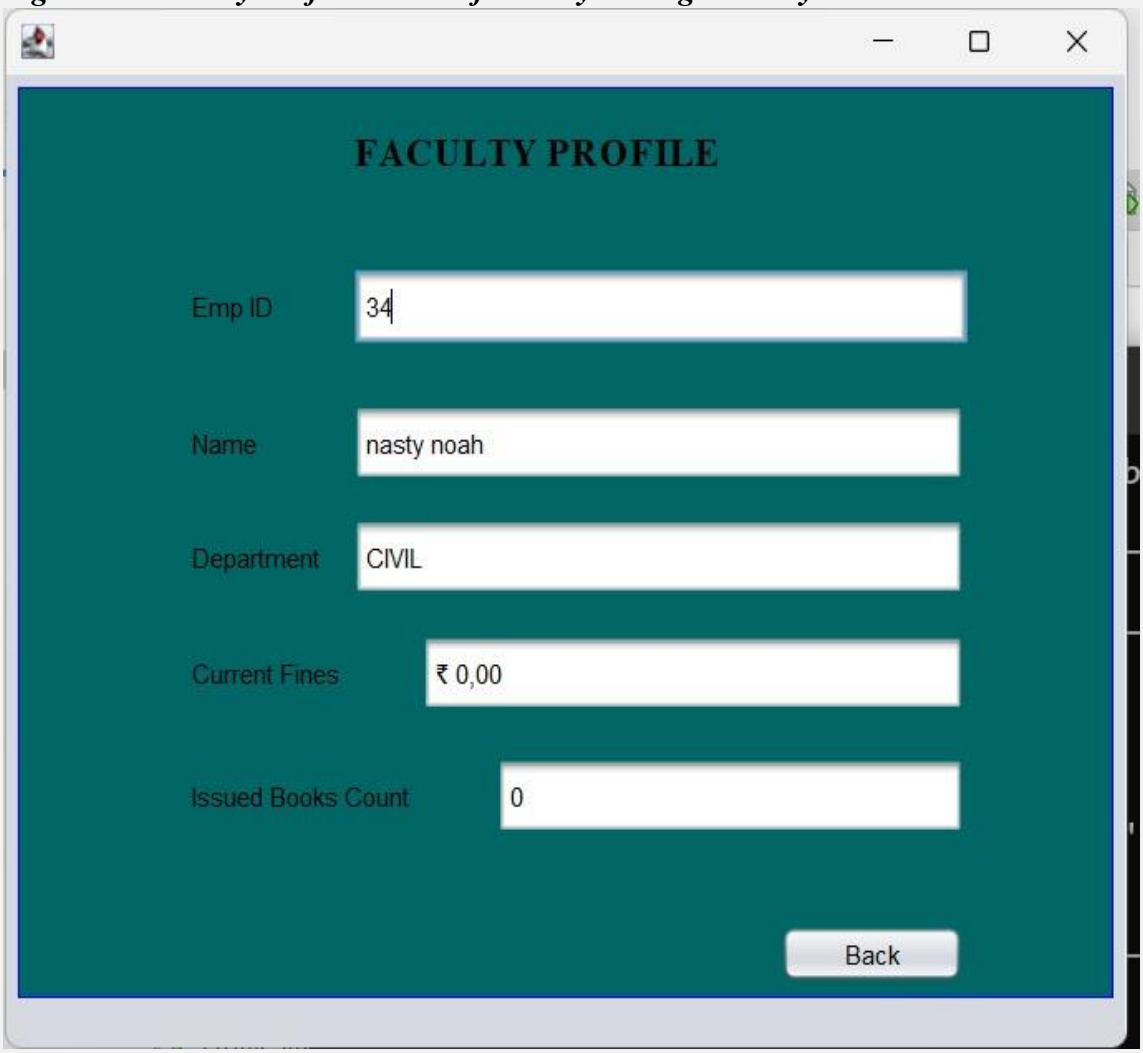
12. Student Profile Screen

Figure 12: Student Profile Screen of Library Management System



13. Faculty Profile Screen

Figure 13: Faculty Profile Screen of Library Management System



7. CONCLUSION

The Library Management System has been successfully developed to manage library operations in an efficient and organized manner. The system replaces manual record-keeping with a computerized solution, thereby reducing errors, saving time, and improving data accuracy.

The application provides essential functionalities such as adding, updating, deleting, and searching books and member records through a user-friendly interface. It also allows issuing and returning of books with proper tracking. By using technologies like Java, Swing, JDBC, and Oracle Database, the system ensures reliable performance, secure data handling, and easy access to library information.

Overall, the system improves the efficiency of managing library resources and reduces the workload of librarians. It also provides a scalable solution that can be further enhanced in the future by adding features such as online access, digital libraries, and automated notifications.

In conclusion, the Library Management System meets the project objectives and serves as an effective tool for managing library activities in educational institutions.

8. FUTURE ENHANCEMENTS

The Library Management System developed in this project provides basic functionalities for managing library records. However, the system can be further enhanced by adding advanced features to improve its functionality and usability.

Some possible future enhancements include:
• Digital Library Integration: Providing access to e-books and online resources
• Fine Management System: Automatic calculation of late return penalties
• Online Book Reservation: Allowing users to reserve books remotely
• Multi-User Support: Enabling multiple users (admin, librarian, students)
• Web-Based Application: Converting the system into a web-based platform
• Notification System: Sending reminders for due dates via email/SMS

These enhancements can make the system more scalable, efficient, and suitable for real-time library management.

9. REFERENCES

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APPENDIX

A.1 Sample Code

Database Connection Code (DBConnection.java):

```
import java.sql.*;
public class DBConnection {
    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("oracle.jdbc.driver.OracleDriver");
            con = DriverManager.getConnection(
                "jdbc:oracle:thin:@localhost:1521:xe",
                "system",
                "password"
            );
        } catch (Exception e) {
            System.out.println(e);
        }
        return con;
    }
}
```

Insert Book Record (InsertBook.java):

```
import java.sql.*;
public class InsertBook {
    public static void main(String[] args) {
        try {
            Connection con = DBConnection.getConnection();
            String query = "INSERT INTO book VALUES(?,?,?,?,?)";
            PreparedStatement pst = con.prepareStatement(query);
```

```
            pst.setInt(1, 101);
            pst.setString(2, "Java Programming");
            pst.setString(3, "Herbert Schildt");
            pst.setString(4, "Programming");
            pst.setInt(5, 5);
            int i = pst.executeUpdate();
            if (i > 0)
                System.out.println("Book Added Successfully");
            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

A.2 SQL Queries

Create Book Table:

```
CREATE TABLE book (
    book_id NUMBER PRIMARY KEY,
    title VARCHAR2(100) NOT NULL,
    author VARCHAR2(100),
    category VARCHAR2(50),
    quantity NUMBER
);
```

Create Member Table:

```
CREATE TABLE member (
    member_id NUMBER PRIMARY KEY,
    member_name VARCHAR2(100) NOT NULL,
```

```
phone VARCHAR2(15),
    email VARCHAR2(100)
);
```

Create Issue Table:

```
CREATE TABLE issue (
    issue_id NUMBER PRIMARY KEY,
    book_id NUMBER,
    member_id NUMBER,
    issue_date DATE,
    return_date DATE,
    FOREIGN KEY (book_id) REFERENCES book(book_id),
    FOREIGN KEY (member_id) REFERENCES member(member_id)
);
```